

Measuring Student Performance

Libraries

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts

## Warning: package 'leaps' was built under R version 4.4.3
```

Dataset

```
## Rows: 1000 Columns: 8
## -- Column specification -----
## Delimiter: ","
## chr (5): gender, race/ethnicity, parental level of education, lunch, test pr...
## dbl (3): math score, reading score, writing score
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

## # A tibble: 6 x 8
##   gender 'race/ethnicity' parental level of educa~1 lunch test preparation cou~2
##   <chr> <chr>           <chr>           <chr> <chr>
## 1 female group B      bachelor's degree stan~ none
## 2 female group C      some college      stan~ completed
## 3 female group B      master's degree   stan~ none
## 4 male group A        associate's degree free~ none
## 5 male group C        some college      stan~ none
## 6 female group B      associate's degree stan~ none
## # i abbreviated names: 1: 'parental level of education',
```

```
## # 2: 'test preparation course'
## # i 3 more variables: 'math score' <dbl>, 'reading score' <dbl>,
## # 'writing score' <dbl>
```

Analysis

Intro

How much does a student's upbringing affect their test scores? Does their race matter? Gender? Is it possible that these factors might be intertwined in affecting a student's result? This analysis hopes to answer some of those questions. The Student Performance data set was found on Kaggle and does not include any deeper explanations on the included variables (what race "groups" are, what the test preparation consisted of, etc).

Data Exploration

One-Hot Encoding categorical variables:

```
## genderfemale gendermale raceA raceB raceC raceD raceE parentEduAssoc
## 1 1 0 0 1 0 0 0 0
## 2 1 0 0 0 1 0 0 0
## 3 1 0 0 1 0 0 0 0
## 4 0 1 1 0 0 0 0 1
## 5 0 1 0 0 1 0 0 0
## 6 1 0 0 1 0 0 0 1
## parentEduBach parentEduHS parentEduMaster parentEduSomeCollege
## 1 1 0 0 0
## 2 0 0 0 1
## 3 0 0 1 0
## 4 0 0 0 0
## 5 0 0 0 1
## 6 0 0 0 0
## parentEduSomeHS lunchFree lunchstandard testPrepcompleted testPrepnone
## 1 0 0 1 0 1
## 2 0 0 1 1 0
## 3 0 0 1 0 1
## 4 0 1 0 0 1
## 5 0 0 1 0 1
## 6 0 0 1 0 1
## mathScore readingScore writingScore
## 1 72 72 74
## 2 69 90 88
## 3 90 95 93
```

```
## 4      47      57      44
## 5      76      78      75
## 6      71      83      78
```

Checking for NA values:

```
## [1] 0 0
```

The size of the dataframe with NA values is [0,0] so there are no NA values.

```
test = c()
a = 1
for (i in 1:nrow(abs(cor(performance_df)))) {
  for (j in 1:ncol(abs(cor(performance_df)))) {
    if (abs(cor(performance_df))[i, j] <= .6) {
      test[a] == abs(cor(performance_df))[i, j]
      a = a+1
    }
  }
}
test
```

```
## NULL
```

Full Model

Full model, reference group not shown in output

```
##
## Call:
## lm(formula = mathScore ~ ., data = performance)
##
## Coefficients:
##              (Intercept)                gendermale
##                -11.6045                 13.2405
##             racegroup B                racegroup C
##                0.8354                 0.1782
##             racegroup D                racegroup E
##                0.0984                 5.0777
## parentEdubachelor's degree      parentEduhigh school
##                -1.0469                 0.5677
##   parentEdumaster's degree      parentEdusome college
##                -1.8561                 0.4003
```

```
## parentEdusome high school          lunchstandard
##                0.5522                3.2127
##                testPrepnone          readingScore
##                3.5023                0.2635
##                writingScore
##                0.7016
```

Best subsets of model:

```
a= regsubsets(mathScore~., data = performance)
rs = summary(a)
rs$which
```

```
## (Intercept) gendermale racegroup B racegroup C racegroup D racegroup E
## 1          TRUE          FALSE          FALSE          FALSE          FALSE          FALSE
## 2          TRUE          TRUE           FALSE          FALSE          FALSE          FALSE
## 3          TRUE          TRUE           FALSE          FALSE          FALSE          FALSE
## 4          TRUE          TRUE           FALSE          FALSE          FALSE          TRUE
## 5          TRUE          TRUE           FALSE          FALSE          FALSE          TRUE
## 6          TRUE          TRUE           FALSE          FALSE          FALSE          TRUE
## 7          TRUE          TRUE           FALSE          FALSE          FALSE          TRUE
## 8          TRUE          TRUE           FALSE          FALSE          FALSE          TRUE
## parentEdubachelor's degree parentEduhigh school parentEdumaster's degree
## 1                                FALSE                                FALSE                                FALSE
## 2                                FALSE                                FALSE                                FALSE
## 3                                FALSE                                FALSE                                FALSE
## 4                                FALSE                                FALSE                                FALSE
## 5                                FALSE                                FALSE                                FALSE
## 6                                FALSE                                FALSE                                FALSE
## 7                                FALSE                                FALSE                                TRUE
## 8                                TRUE                                 FALSE                                TRUE
## parentEdusome college parentEdusome high school lunchstandard testPrepnone
## 1                                FALSE                                FALSE                                FALSE                                FALSE
## 2                                FALSE                                FALSE                                FALSE                                FALSE
## 3                                FALSE                                FALSE                                FALSE                                TRUE
## 4                                FALSE                                FALSE                                FALSE                                TRUE
## 5                                FALSE                                FALSE                                TRUE                                 TRUE
## 6                                FALSE                                FALSE                                TRUE                                 TRUE
## 7                                FALSE                                FALSE                                TRUE                                 TRUE
## 8                                FALSE                                FALSE                                TRUE                                 TRUE
## readingScore writingScore
## 1          TRUE          FALSE
## 2          FALSE          TRUE
## 3          FALSE          TRUE
```

```
## 4      FALSE      TRUE
## 5      FALSE      TRUE
## 6      TRUE       TRUE
## 7      TRUE       TRUE
## 8      TRUE       TRUE
```

```
step(g, direction = "both")
```

```
## Start:  AIC=3373.54
## mathScore ~ gender + race + parentEdu + lunch + testPrep + readingScore +
##   writingScore
##
##           Df Sum of Sq  RSS   AIC
## <none>                28319 3373.5
## - parentEdu      5      441 28761 3379.0
## - readingScore   1     1129 29448 3410.6
## - lunch          1     2119 30438 3443.7
## - testPrep       1     2242 30561 3447.7
## - race           4     2779 31098 3459.1
## - writingScore    1     7471 35790 3605.7
## - gender         1    36436 64755 4198.6

##
## Call:
## lm(formula = mathScore ~ gender + race + parentEdu + lunch +
##   testPrep + readingScore + writingScore, data = performance)
##
## Coefficients:
##           (Intercept)                gendermale
##           -11.6045                13.2405
##           racegroup B                racegroup C
##             0.8354                0.1782
##           racegroup D                racegroup E
##             0.0984                5.0777
## parentEdubachelor's degree      parentEduhigh school
##           -1.0469                0.5677
##   parentEdumaster's degree      parentEdusome college
##           -1.8561                0.4003
##   parentEdusome high school      lunchstandard
##             0.5522                3.2127
##           testPrepnone            readingScore
##             3.5023                0.2635
##           writingScore
##             0.7016
```